

Theory note S1 — Frequency hopping under partial-band jamming: processing gain, the worst-case band fraction, and the jamming margin

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Purpose. S1 adds a hopping transmitter and a partial-band noise jammer to the S0 link. This note derives (i) why hopping buys a processing gain, (ii) the closed-form BER used to verify the simulator, (iii) the jammer’s optimal band fraction ρ^* and the resulting inverse-linear BER law, (iv) the band fraction that matters for a link *outage*, which S2 builds on, and (v) the jamming margin numbers reported in `results/s1_link_budget.txt`. The worst-case analysis in Section 3 is the standard textbook treatment of pulsed and partial-band noise jamming [1, 2]; it is reproduced here so that every number in the code can be traced to an equation. Mathematics used: total probability, one-variable optimisation.

1 Processing gain

The data occupies bandwidth W_d ; the transmitter hops among N_{ch} channels spread over $W_{ss} \approx N_{\text{ch}}W_d$. A jammer with total power J that wants to cover the whole hopping band must spread J over W_{ss} , so its density is $N_J = J/W_{ss}$. The transmitter’s signal at any instant occupies only W_d , where the jammer delivers only $N_JW_d = JW_d/W_{ss} = J/N_{\text{ch}}$ of its power. The jammer effectively loses a factor

$$G_p = \frac{W_{ss}}{W_d} \approx N_{\text{ch}}, \quad G_p [\text{dB}] = 10 \log_{10} N_{\text{ch}}.$$

With $N_{\text{ch}} = 100$: $G_p = 20 \text{ dB}$.

Relating E_b/N_J to J/S . With signal power S and bit rate R_b , $E_b = S/R_b$. Taking $W_d \approx R_b$ (a different bandwidth convention changes G_p by a constant and shifts every J/S axis by the same amount),

$$\frac{E_b}{N_J} = \frac{S/R_b}{J/W_{ss}} = \frac{S}{J} \cdot \frac{W_{ss}}{R_b} \approx \frac{S}{J} G_p \quad \implies \quad \boxed{\frac{E_b}{N_J} [\text{dB}] = G_p [\text{dB}] - \frac{J}{S} [\text{dB}] }.$$

This is the line `ebnj_lin = db_to_linear(gp_db - js_db)` in the code and the reason the figure’s x -axis is J/S while the analysis runs on E_b/N_J .

2 Partial-band jamming: the mixture formula

A jammer need not spread over the whole band. It can concentrate J on a fraction $\rho \in (0, 1]$ of the channels, raising its density there to N_J/ρ and leaving the rest clean. The jammer does not know

the pseudo-random hopping pattern, so each hop lands in the jammed region independently with probability ρ . By the law of total probability,

$$P_b(\rho) = (1 - \rho) Q\left(\sqrt{\frac{2E_b}{N_0}}\right) + \rho Q\left(\sqrt{\frac{2E_b}{N_0 + N_J/\rho}}\right). \quad (1)$$

The first term is the thermal-noise-only BER of S0 (clean hop); the second is an S0 link whose noise density is N_0 plus the concentrated jammer density. This is `ber_partial_band_theory()` and it is what the Monte Carlo (random jammed flag per hop, then the S0 signal model) is checked against, point by point, with the z -score test of the S0 note (`VERIFICATION 1`).

Hop rate. The simulation sends one bit per hop so that successive bits are jammed independently. Eq. (1) is an *average* over hops and holds for any number of bits per hop; what changes with slow hopping is the burstiness of the errors, which matters for coding and is outside this model.

Limit check ($\rho = 1$). Eq. (1) reduces to $Q(\sqrt{2E_b/(N_0 + N_J)})$: plain AWGN with the two densities added (`VERIFICATION 2`, agreement to 10^{-12}).

3 The jammer's optimal ρ and the inverse-linear law

From the jammer's point of view ρ is a free parameter and it wants to maximise (1). When the jammer dominates the thermal noise ($N_J/\rho \gg N_0$), drop N_0 :

$$P_b(\rho) \approx \rho Q\left(\sqrt{\frac{2\rho E_b}{N_J}}\right).$$

Substitute $x = 2\rho E_b/N_J$ (so $\rho = x/(2E_b/N_J)$):

$$P_b \approx \frac{x Q(\sqrt{x})}{2 E_b/N_J} = \frac{g(x)}{2 E_b/N_J}, \quad g(x) := x Q(\sqrt{x}).$$

The dependence on E_b/N_J has been pulled out; the jammer just maximises $g(x)$ over $x > 0$. Setting $g'(x) = 0$, with $Q'(u) = -\varphi(u)$ and $\varphi(u) = e^{-u^2/2}/\sqrt{2\pi}$,

$$g'(x) = Q(\sqrt{x}) - x \varphi(\sqrt{x}) \frac{1}{2\sqrt{x}} = Q(\sqrt{x}) - \frac{\sqrt{x}}{2} \varphi(\sqrt{x}) = 0.$$

This transcendental equation has one positive root, found numerically:

$$x^* \approx 1.418, \quad g(x^*) \approx 0.1657, \quad Q(\sqrt{x^*}) \approx 0.1169.$$

Numerical check: at $x = 1.418$ both sides of $Q(\sqrt{x}) = \frac{\sqrt{x}}{2} \varphi(\sqrt{x})$ equal 0.1169.

Two consequences follow.

$$\rho^* = \frac{x^*}{2 E_b/N_J} \approx \frac{0.709}{E_b/N_J}, \quad (2)$$

$$P_b^{\text{worst}} = \frac{g(x^*)}{2 E_b/N_J} \approx \frac{0.083}{E_b/N_J}. \quad (3)$$

Eq. (3) is the main result of S1: *against an optimal partial-band jammer the BER decays only as $1/(E_b/N_J)$* , not exponentially as in S0. (Eq. (2) is only meaningful while $\rho^* \leq 1$, i.e. $E_b/N_J \geq x^*/2 \approx -1.5$ dB; below that the jammer uses the whole band.) These constants are the ones quoted for coherent BPSK in [1]. For noncoherent BFSK, the modulation more commonly paired with frequency hopping, the same argument gives $\rho^* = 2/(E_b/N_J)$ and $P_b^{\text{worst}} = e^{-1}/(E_b/N_J)$ [1, 2]: a different constant, the same inverse-linear law. Coherent BPSK is used here only because it continues the S0 model.

Check of the optimiser. The script finds ρ^* with a bounded one-dimensional search on the exact formula (1). With thermal noise switched off it must return the constants above; VERIFICATION 3 obtains $\rho^* E_b/N_J = 0.7088$ and $P_b^{\text{worst}} E_b/N_J = 0.08286$.

The number in the figure. At $J/S = 12$ dB with $G_p = 20$ dB, $E_b/N_J = 8$ dB = 6.31. Eq. (2) gives $\rho^* = 0.709/6.31 = 0.112$ with thermal noise ignored. Keeping the thermal term N_0 in (1) shifts the optimum upward; the search on the exact formula gives $\rho^* = 0.135$. The curve is flat near its maximum (the BER at $\rho = 0.10$ and at $\rho = 0.16$ is within 3 % of the peak), so the Monte Carlo points are used to confirm the *curve*, not to locate the maximum. The worst-case BER is 1.42×10^{-2} versus 2.70×10^{-3} for a full-band jammer of the same power, a factor of 5.2.

4 The band fraction that causes an outage

S2 declares a link *down* when $P_b > \text{BER}_{\text{out}}$. The jammer then asks a different question: not “which ρ maximises the BER of this link” but “what is the strongest link I can still bring down”. By Eq. (3) the worst-case BER reaches BER_{out} when

$$(E_b/N_J)_{\text{out}} = \frac{0.0829}{\text{BER}_{\text{out}}}, \quad \rho_{\text{out}} = \frac{0.709}{(E_b/N_J)_{\text{out}}} = \frac{\text{BER}_{\text{out}}}{Q(\sqrt{x^*})} \approx 8.55 \text{BER}_{\text{out}}. \quad (4)$$

For $\text{BER}_{\text{out}} = 10^{-3}$: $(E_b/N_J)_{\text{out}} = 82.9 = 19.2$ dB and $\rho_{\text{out}} = 0.0086$. This is Eq. (2) evaluated at the outage point, not a new optimisation; it is much smaller than the 0.135 of the figure only because 19.2 dB is a much larger E_b/N_J than 8 dB.

One-channel limit. A jammer facing N_{ch} channels cannot jam less than one of them, so $\rho \geq 1/N_{\text{ch}} = 0.01$. Here $\rho_{\text{out}} = 0.0086$ is just below that limit. The script optimises ρ as a continuous variable, which keeps the link to Eqs. (2)–(3), and reports the one-channel case alongside; the two differ by 0.01 dB (next section).

5 Jamming margin

A link is *up* if $P_b \leq \text{BER}_{\text{out}} = 10^{-3}$. The value is an assumption; the S2 note shows that changing it shifts every threshold by 10 dB per decade and leaves the dependence on network size unchanged. Solve the BER formula for the E_b/N_J at which the criterion is just met, $(E_b/N_J)_{\text{req}}$, then allow an implementation loss L_{sys} (assumed 2 dB; it shifts all margins equally). The jamming margin is the J/S the link can absorb:

$$M_J [\text{dB}] = G_p - (E_b/N_J)_{\text{req}} - L_{\text{sys}}.$$

For $N_{\text{ch}} = 100$, $E_b/N_0 = 10$ dB, $L_{\text{sys}} = 2$ dB:

jammer strategy	optimum ρ	$(E_b/N_J)_{\text{req}}$	M_J
full band, $\rho = 1$	—	9.61 dB	8.39 dB
optimal ρ (continuous)	0.0095	19.54 dB	− 1.54 dB
optimal $\rho \geq 1/N_{\text{ch}}$	0.0100	19.53 dB	− 1.53 dB

The required 19.54 dB is slightly above the 19.2 dB of Eq. (4) because the thermal noise ($E_b/N_0 = 10$ dB) is not negligible inside the jammed channels. The optimal jammer needs 9.9 dB *less* power than a full-band jammer to break the link. The negative margin means that, at $\text{BER}_{\text{out}} = 10^{-3}$ and without coding, 100 hop channels do not protect a BPSK link against a jammer of equal received power.

6 Correspondence with the code

Mathematics	Code
$G_p = 10 \log_{10} N_{\text{ch}}$	<code>gp_db = linear_to_db(N_CH)</code>
$E_b/N_J = G_p - J/S$	<code>ebnj_lin = db_to_linear(gp_db - js_db)</code>
jammed hop w.p. ρ	<code>jammed = rng.random(...) < rho</code>
$\sigma^2 = (N_0 + N_J/\rho)/2$	<code>sigma_jammed = np.sqrt((n0 + nj / rho) / 2.0)</code>
Eq. (1)	<code>ber_partial_band_theory()</code>
ρ^* on the exact formula	<code>worst_case_rho()</code>
$(E_b/N_J)_{\text{req}}$	<code>required_ebnj_db()</code> (root of $P_b - \text{BER}_{\text{out}}$)
$1/N_{\text{ch}}$ limit	<code>RHO_MIN_PHYSICAL</code>

References

- [1] B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed. Prentice Hall, 2001, ch. 12 (spread-spectrum techniques; jamming considerations).
- [2] M. K. Simon, J. K. Omura, R. A. Scholtz, and B. K. Levitt, *Spread Spectrum Communications Handbook*, rev. ed. McGraw-Hill, 1994.